

OSCAT
Revision History

Release	Modul name	Rev	Date	History
271	AOUT	1.4	23.02.08	Improved code for higher speed
271	AOUT1	1.0	23.02.08	New Module
271	AIN1	1.0	23.02.08	New Module
271	POLYNOM_INT	1.2	22.02.08	Improved code for higher speed
271	TRIANGLE_A	1.1	22.02.08	Improved code for higher speed
271	SPHERE_V	1.2	22.02.08	Improved code for higher speed
271	CONE_V	1.2	22.02.08	Improved code for higher speed
271	CIRCLE_C	1.1	22.02.08	Improved code for higher speed
271	CIRCLE_A	1.2	22.02.08	Improved code for higher speed
271	CABS	1.0	21.02.08	New Module
271	CACOS	1.0	21.02.08	New Module
271	CACOSH	1.0	21.02.08	New Module
271	CADD	1.0	21.02.08	New Module
271	CARG	1.0	21.02.08	New Module
271	CASIN	1.0	21.02.08	New Module
271	CASINH	1.0	21.02.08	New Module
271	CATAN	1.0	21.02.08	New Module
271	CATANH	1.0	21.02.08	New Module
271	CCON	1.0	21.02.08	New Module
271	CCOS	1.0	21.02.08	New Module
271	CCOSH	1.0	21.02.08	New Module
271	CDIV	1.0	21.02.08	New Module
271	CEXP	1.0	21.02.08	New Module
271	CINV	1.0	21.02.08	New Module
271	CLOG	1.0	21.02.08	New Module
271	CMUL	1.0	21.02.08	New Module
271	CPOL	1.0	21.02.08	New Module
271	CPOW	1.0	21.02.08	New Module
271	CSET	1.0	21.02.08	New Module
271	CSIN	1.0	21.02.08	New Module
271	CSINH	1.0	21.02.08	New Module
271	CSQRT	1.0	21.02.08	New Module
271	CSUB	1.0	21.02.08	New Module
271	CTAN	1.0	21.02.08	New Module
271	CTANH	1.0	21.02.08	New Module
271	HYPOT	1.0	21.02.08	New Module
271	TEMPERATURE	1.0	21.02.08	New Module
271	RTC_2	1.1	20.02.08	Added millisecond set input
271	RTC_MS	1.1	20.02.08	Added millisecond set input
271	WORD_OF_BYTE	1.2	19.02.08	replaced + with or for better compatibility
271	D_TRIG	1.1	19.02.08	Improved code for higher speed
271	STACK_32	1.1	19.02.08	Improved code for higher speed
271	STACK_16	1.1	19.02.08	Improved code for higher speed
271	FIFO_32	1.1	19.02.08	Improved code for higher speed
271	FIFO_16	1.1	19.02.08	Improved code for higher speed
271	PARITY	1.3	18.02.08	Improved code for higher speed
271	CHECK_PARITY	1.3	18.02.08	Improved code for higher speed
271	BYTE_OF_BIT	1.1	18.02.08	Improved code for higher speed
271	REVERSE	1.1	18.02.08	Improved code for higher speed
271	BLIND_SECURITY	1.2	12.02.08	added config variables rain_up and wind_up position for rain and wind is now configurable
27	FT_INT	1.7	08.02.08	Added interim variable to allow for changes smaller than 8 digits of output
27	METER	1.1	08.02.08	Use new version of ft_int and deleted unnecessary limits for integrators
27	SWAP_BYTE2	1.0	04.02.08	New Module
27	SWAP_BYTE	1.0	04.02.08	New Module
27	REPLACE_ALL	1.0	04.02.08	New Module
27	CHR	1.1	04.02.08	return string would not be terminated properly
27	FINDP	1.0	04.02.08	New Module
27	MIRROR	1.0	04.02.08	New Module
27	TRIM1	1.0	04.02.08	New Module
27	LOWERCASE	1.1	04.02.08	improved performance added support for Ä, Ö, Ü

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27	UPPERCASE	1.1	04.02.08	improved performance
27	EASTER	1.2	03.02.08	added support for Å, Ö, Ü
27	RDM	1.4	02.02.08	Modified code for better performance
27	BLIND_SCENE	1.1	02.02.08	changed algorithm to avoid non iec functions and guarantee more randomness
				Replaced shr/shl with and for better performance
27	FT_RMP	1.4	25.01.08	Improved algorithm for higher speed
27	DCF77	1.7	23.01.08	allow KR and KF to be 0
27	RTC_2	1.0	21.01.08	added sdt input which is used to initialize rtc and rtc1 during first cycle.
27	RTC_MS	1.0	20.01.08	New module
				New Module
27	SUN_POS	1.2	18.01.08	further performance improvements
27	SUN_TIME		17.01.08	only calculate once every 10 seconds
				calculation is now only performed once a day
				New code with better performance
27	TIMER_1	1.6	17.01.08	Output is now only active during the time of day it is supposed to
27	COUNT_BR	1.0	16.01.08	no pulse is generated at power up
27	SELECT_8	1.0	16.01.08	New Module
27	INC	1.1	15.01.08	New Module
27	RANGE_TO_WORD	1.0	09.01.08	Allow for negative increment
27	RANGE_TO_BYTE	1.0	09.01.08	New Module
27	WORD_TO_RANGE	1.0	09.01.08	New Module
27	BYTE_TO_RANGE	1.0	09.01.08	New Module
27	DEW_TEMP	1.4	08.01.08	Improved algorithm for higher speed
27	LEAP_OF_DATE	1.2	08.01.08	Improved algorithm for higher speed
27	DAY_OF_MONTH	1.4	08.01.08	Improved algorithm for higher speed
27	COTH	1.2	08.01.08	Improved algorithm for higher speed
27	ROUND	1.2	08.01.08	Improved algorithm for higher speed
27	SH_2	1.4	06.01.08	Improved algorithm for higher speed
27	STAIR	1.2	06.01.08	Improved algorithm for higher speed
27	SCALE_B8	1.2	06.01.08	Improved algorithm for higher speed
27	SCALE_B4	1.2	06.01.08	Improved algorithm for higher speed
27	SCALE_B2	1.2	06.01.08	Improved algorithm for higher speed
27	GEN_SQR	1.3	06.01.08	Improved algorithm for higher speed
27	GEN_SIN	1.3	06.01.08	Improved algorithm for higher speed
27	TEMP_SI	1.4	06.01.08	Improved algorithm for higher speed
27	TEMP_NI	1.4	06.01.08	Improved algorithm for higher speed
27	BAR_GRAPH	1.2	06.01.08	Improved algorithm for higher speed
27	KMH_TO_MS	1.1	06.01.08	Improved algorithm for higher speed
27	F_TO_C	1.1	06.01.08	Improved algorithm for higher speed
27	SUN_POS	1.1	06.01.08	Improved algorithm for higher speed
27	SUN_TIME	1.3	06.01.08	Improved algorithm for higher speed
27	_ARRAY_SORT	1.5	05.01.08	Improved algorithm for higher speed
27	SINH	1.2	05.01.08	Improved algorithm for higher speed
27	COSH	1.2	05.01.08	Improved algorithm for higher speed
27	ATANH	1.1	05.01.08	Improved algorithm for higher speed
27	ACOTH	1.1	05.01.08	Improved algorithm for higher speed
27	HYST_3	1.2	05.01.08	Improved code for higher performance
27	HYST_2	1.1	05.01.08	Improved code for higher performance
27	FT_PT1	1.7	05.01.08	Improved code for higher performance
27	FT_PID	1.6	05.01.08	Improved code for higher performance
27	FT_PI	1.2	05.01.08	Improved performance
27	FT_INT	1.6	05.01.08	Improved performance
27	TEMP_PT	1.5	05.01.08	replaced /2 with * 0.5 for better performance
27	RDM	1.3	05.01.08	Corrected an error with high values of sps timer
27	PERIOD	1.2	04.01.08	Improved performance
27	DAY_OF_YEAR	1.2	04.01.08	Improved performance
27	YEAR_BEGIN	1.1	04.01.08	formula for step7 was incorrect during leap years
27	SET_DATE	1.4	03.01.08	Improved performance
27	IRTRANS_DECODE	1.0	02.01.08	New Module
27	IRTRANS_8	1.0	02.01.08	New Module
27	IRTRANS_4	1.0	02.01.08	New Module
27	IRTRANS_1	1.0	02.01.08	New Module
27	DRIVER_4	1.0	02.01.08	New Module
27	DRIVER_1	1.0	02.01.08	New Module

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27	CLK_DIV	1.1	02.01.08	Improved performance
27	BYTE_OF_DWORD	1.1	02.01.08	Improved performance
27	WORD_OF_DWORD	1.1	02.01.08	Improved performance
27	FIB	1.1	02.01.08	deleted unused variable pt
27	WORD_OF_BYTE	1.1	02.01.08	Improved performance
27	CRC_GEN	1.3	02.01.08	Improved performance
27	DWORD_OF_BYTE	1.1	02.01.08	Improved performance
26	BLIND_INPUT	1.2	30.12.07	added click_timeout variable and function
26	FIB	1.0	27.12.07	New Module
26	WATER_CP	1.1	27.12.07	Increased accuracy
26	SHR1	1.1	27.12.07	changed code for better performance
26	SHL1	1.1	27.12.07	changed code for better performance
26	LTCH_4	1.1	27.12.07	changed code for better performance
26	LTCH	1.1	27.12.07	changed code for better performance
26	FF_JKE	1.1	27.12.07	changed code for better performance
26	FF_DRE	1.1	27.12.07	changed code for better performance
26	FF_D4E	1.1	27.12.07	changed code for better performance
26	FF_D2E	1.1	27.12.07	changed code for better performance
26	STAIR	1.1	27.12.07	changed code for better performance
26	STAIR2	1.1	27.12.07	changed code for better performance
26	HYST_3	1.1	27.12.07	changed code for better performance
26	ACTUATOR_2P	1.5	27.12.07	changed code for better performance
26	LINEAR_INT	1.1	27.12.07	changed code for better performance
26	STORE_8	1.0	26.12.07	New Module
26	MESSAGE_8	1.0	26.12.07	New Module
26	FADE	1.0	26.12.07	New Module
26	RMP_B	1.6	25.12.07	changed code for better performance
26	RMP_W	1.7	25.12.07	corrected an error while step response was too slow for fast rist times
26	HEAT_METER	1.1	25.12.07	added config variable return_meter
26	MATRIX	1.2	23.12.07	added output C for current consumption
26	WEEKDAY	1.2	23.12.07	added exit statement in for loop instead of i:=5
26	TEMP_PT	1.2	23.12.07	correction for step7
26	YEAR	1.4	23.12.07	avoid a negative square root if input values are wrong
26	WORK_WEEK	1.2	23.12.07	changed code for better performance
26	YEAR_BEGIN	1.2	19.12.07	changed code for better performance
26	TEMP_NI	1.0	19.12.07	New Module
26	POLYNOM_INT	1.3	17.12.07	changed code for better performance
26	TEMP_SI	1.1	17.12.07	init makes no sense for a function
26	YEAR	1.3	17.12.07	changed code for better performance
26	SET_DATE	1.2	16.12.07	changed code for better performance
26	OSCAT_VERSION	1.3	16.12.07	changed code for better performance
26	CLK_8MS	1.1	16.12.07	added possibility to return date and version deponding on in
26	CLK_4MS		16.12.07	No longer supported is replaced by CLK_N
26	CLK_2MS		16.12.07	No longer supported is replaced by CLK_N
26	CLK_1MS		16.12.07	No longer supported is replaced by CLK_N
26	CLK_N	1.0	16.12.07	New Module
26	EASTER	1.1	15.12.07	changed code for better performance
26	REAL_TO_STRF	1.2	15.12.07	changed code for better performance
26	DWORD_TO_STRH	1.2	15.12.07	changed code for better performance
26	DWORD_TO_STRB	1.1	15.12.07	changed code for better performance
26	BYTE_TO_STRH	1.2	15.12.07	changed code for better performance
26	BYTE_TO_STRB	1.1	15.12.07	changed code for better performance
26	FINDB	1.1	15.12.07	changed code for better performance
26	TICKER	1.1	15.12.07	step now starts at 1 instaed of 0
26	_ARRAY_SORT	1.4	15.12.07	added function return = true
26	SIGNAL_4	1.0	13.12.07	New Module
26	SIGNAL	1.0	13.12.07	New Module
26	INT_TO_BCD	1.0	13.12.07	New Module
26	BCD_TO_INT	1.0	13.12.07	New Module
26	ARRAY_VAR	1.1	12.12.07	changed code for better performance
26	ARRAY_AVG	1.1	12.12.07	changed code for better performance
26	ARRAY_VAR	1.1	12.12.07	changed code for better performance
26	ARRAY_TREND	1.1	12.12.07	changed code for better performance
26	V3_ZANG	1.0	11.12.07	New Module

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26	V3_YANG	1.0	11.12.07	New Module
26	V3_XANG	1.0	11.12.07	New Module
26	V3_XPRO	1.0	11.12.07	New Module
26	V3_SMUL	1.0	11.12.07	New Module
26	V3_REV	1.0	11.12.07	New Module
26	V3_PAR	1.0	11.12.07	New Module
26	V3_NORM	1.0	11.12.07	New Module
26	V3_DPRO	1.0	11.12.07	New Module
26	V3_SUB	1.0	11.12.07	New Module
26	V3_ADD	1.0	11.12.07	New Module
26	V3_ANG	1.0	11.12.07	New Module
26	V3_ABS	1.0	11.12.07	New Module
26	LANGEVIN	1.0	10.12.07	New Module
26	SIGMOID	1.0	10.12.07	New Module
26	GAMMA	1.0	10.12.07	New Module
26	FACT	1.2	10.12.07	start value for i has changed to 2 for better performance
26	TONOF	1.0	10.12.07	New Module
26	PWM_PW	1.3	09.12.07	Corrected a problem with F = 0
252	DCF77	1.6	08.12.07	corrected a problem with time_zone_offset
251	LTIME_TO_UTC	1.3	08.12.07	corrected a problem with time_zone_offset
251	LTIME	1.6	08.12.07	corrected a problem with time_zone_offset
251	LOCAL_TIME	1.4	08.12.07	corrected a problem with time_zone_offset
25	ESR_COLLECT	1.1	08.12.07	Added reset input
25	ESR_MON_R4	1.2	08.12.07	corrected an error while esr type would not be set
25	CHECK_PARITY	1.2	08.12.07	changed code for better performance
25	PARITY	1.2	08.12.07	changed code for better performance
25	WCT	1.1	07.12.07	changed code for better performance
25	SPHERE_V	1.1	04.12.07	changed code for better performance
25	CONE_V	1.1	04.12.07	changed code for better performance
25	CIRCLE_A	1.1	04.12.07	changed code for better performance
25	EXP_N	1.0	04.12.07	New Module
25	TICKER	1.0	04.12.07	New Module
25	AIR_DENSITY	1.1	04.12.07	changed code for better performance
25	SQRTN	1.1	02.12.07	changed code for better performance
25	ROUND	1.1	02.12.07	changed code for better performance
25	ASINH	1.1	02.12.07	changed code for better performance
25	ACOSH	1.1	02.12.07	changed code for better performance
25	AOUT	1.3	02.12.07	changed code for better performance
25	SDD	1.1	02.12.07	changed code for better performance
25	DEW_TEMP	1.3	02.12.07	changed code for better performance
25	DEW_CON	1.2	02.12.07	changed code for better performance
25	AIR_ENTHALPY	1.2	02.12.07	output is now in Gramms / Qubic meter
25	EXP10	1.0	02.12.07	changed code for better performance
25	AIN	1.4	02.12.07	New Module
25	RMP_W	1.6	02.12.07	changed code for better performance
25	TEMP_PT	1.3	02.12.07	corrected an error in calculation of step response
25	SENSOR_INT	1.1	02.12.07	changed code for better performance
25	RES_SI	1.1	02.12.07	Corrected an error in algorithm
25	RES_PT	1.1	02.12.07	changed code for better performance
25	RES_NI	1.2	02.12.07	changed code for better performance
25	ONTIME	1.2	02.12.07	changed code for better performance
25	BAR_GRAPH	1.4	02.12.07	changed code for better performance
25	FT_INT	1.1	02.12.07	changed code to improove performance
25	WATER_DENSITY	1.5	02.12.07	changed code to improove performance
25	HEAT_INDEX	1.1	02.12.07	new code for better performance
25	DEW_RH	1.2	02.12.07	new code for better performance
25	DST	1.1	02.12.07	new code for better performance
25	COTH	1.2	01.12.07	changed code to improove performance
25	COSH	1.1	01.12.07	changed code to improove performance
25	TANH	1.1	01.12.07	changed code to improove performance
25	SINH	1.1	01.12.07	changed code to improove performance
25	SIGN_R	1.1	01.12.07	changed code to improove performance
25	SIGN_I	1.2	01.12.07	changed code to improove performance
25	EVEN	1.2	01.12.07	changed code to improove performance

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				changed out to be K * in during initialization
25	FT_PT2	1.3	30.11.07	avoid divide by 0 if TN = 0
25	FT_PT1	1.6	30.11.07	changed out to be K * in during initialization
25	PWM_PW	1.2	27.11.07	avoid divide by 0 when F = 0
25	PWM_DC	1.2	27.11.07	avoid divide by 0 when F = 0
25	GEN_SIN	1.2	27.11.07	avoid divide by 0 when pt = 0
25	GEN_RMP	1.2	27.11.07	avoid divide by 0 when pt = 0
24	BLIND_SHADE	1.2	24.11.07	added shade_delay to avoid constant up and down if sun is out partially
24	HOLIDAY	1.3	24.11.07	changes F_use of Buß_und_Bettag to 0 because this is no official holiday
24	REAL_TO_STRF	1.1	20.11.07	when N=0 there will be no dot at the end of the string
24	GEN_RDT	1.0	20.11.07	New Module
24	RDM	1.2	20.11.07	changed code of temp calculation to avoid overflow in modr due to resuclt would not fit
24	ACTUATOR_2P	1.4	19.11.07	DINT for high timer values
24	MIX	1.0	19.11.07	replaced left over statement time() with tx
24	HEX_TO_DWORD	1.1	19.11.07	New Module
24	HEAT_METER	1.0	17.11.07	changed type of function from int to dword
24	_ARRAY_INIT	1.2	14.11.07	New Module
24	_ARRAY_SORT	1.3	14.11.07	changed size calculation to be more efficient
24	_ARRAY_MEDIAN	1.4	14.11.07	changed size calculation to be more efficient
24	T_PLC_MS	1.1	12.11.07	corrected a problem with size calculation
24	T_PLC_US	1.1	12.11.07	added temporaray variable tx because some compilers could not handle time() as an argument
24	LTIME_TO_UTC	1.2	12.11.07	added temporaray variable tx because some compilers could not handle time() as an argument
24	LTIME	1.5	12.11.07	changed Type of time_zone_offset from time to int
24	LOCAL_TIME	1.3	12.11.07	to allow for time zones with negative offset
24	DCF77	1.5	12.11.07	changed Type of time_zone_offset from time to int
24	RDM	1.1	11.11.07	to allow for time zones with negative offset
24	MUL_ADD	1.1	11.11.07	changed time_zone_offset from time to integer to allow for negative offset time zones
24	NEG	1.0	11.11.07	changed time_offset from time to integer to allow for negative offset time zones
24	SGN	1.1	11.11.07	changed time() into t_plc_ms()
24	LTIME_TO_UTC	1.1	05.11.07	deleted preset values for K and O this makes no sense for a function
24	CONTROL_SET2	1.0	04.11.07	New Module
24	CONTROL_SET1	1.0	04.11.07	New Module
24	TUNE	1.0	04.11.07	New Module
24	TUNE2	1.0	03.11.07	New Module
24	PARSET2	1.0	03.11.07	New Module
24	FT_PI	1.1	03.11.07	added noise input to filter noise
24	FT_PID	1.5	03.11.07	output diff added
24	SET_TOD	1.3	02.11.07	set limit output false when output is within limits
24	PARSET	1.0	02.11.07	overfolw was not set correctly
24	DEAD_BAND	1.0	02.11.07	added noise input to filter noise,
24	AIR_ENTHALPY	1.1	02.11.07	output diff added,
23	DATE_ADD	1.2	01.11.07	set limit output false when output is within limits
23	SET_TOD	1.2	01.11.07	overfolw was not set correctly
23	SET_DT	1.4	01.11.07	changed dword to DINT in calcualtion to avoid warnings with some compilers
23	WINDOW2	1.0	31.10.07	New Module
23	FT_PID	1.4	31.10.07	deleted unused variable RF
23	FT_PI	1.0	31.10.07	added int_to_dword stetemets to avoid possible overrun with möller ecp4
23	AIR_ENTHALPY	1.0	31.10.07	changed coding to avoid possible overrun situation on möller ecp4
23	HOLIDAY	1.2	31.10.07	converted hour type integer to dowrd in calculation to avoid overrun on möller ecp4
23	BLIND_CONTROL	1.3	30.10.07	Rewritten to avoid error when one of the limits is 0
23	AIR_DENSITY	1.0	29.10.07	New Module
23	SDD	1.0	29.10.07	New Module
23	WATER_ENTHALPY	1.0	29.10.07	New Module
23	FT_DERIV	1.3	29.10.07	prohibit calculation when tx - last = 0 to avoid div 0 and increase accuracy on fast systems

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23	FT_INT	1.4	29.10.07	changed code so int will not be called is time difference is 0 to increase accuracy on systems with cycle times below 1ms
23	BLIND_SCENE	1.0	24.10.07	new module
23	LTIME	1.4	24.10.07	changed code because the execution every 100ms can cause major problems if the supplied time was not correct at start. use of new dst function
23	LOCAL_TIME	1.2	24.10.07	changed dst calculation because function dst was upgraded no error in local_time only a change in DST
23	DCF77	1.4	24.10.07	changed dst calculation because function dst was upgraded no error in DCF77 only a change in DST
23	DST	1.1	24.10.07	deleted time_zone_offset input because dst is generally at 01:00 utc and not mesz uk starts 01:00 utc and also greece
23	FT_PT1	1.5	23.10.07	added out := in to the init statements
23	BLIND_SHADE	1.1	19.10.07	manual mode was not detected because outputs were checked and not inputs UP and DN
23	CONE_V	1.0	17.10.07	New Module
22	GLOBAL CONSTANTS		16.10.07	Renamed PI in PI1 for better compatibility to other libraries
22	DEG	1.1	16.10.07	Deg will become 0 when it reaches 360
22	RAD	1.1	16.10.07	Rad will become 0 when it reaches 2PI
22	TRIANGLE_A	1.0	16.10.07	New Module
22	SPHERE_V	1.0	16.10.07	New Module
22	ELLIPSE_C	1.0	16.10.07	New Module
22	ELLIPSE_A	1.0	16.10.07	New Module
22	CIRCLE_C	1.0	16.10.07	New Module
22	CIRCLE_A	1.0	16.10.07	New Module
22	SGN	1.0	16.10.07	New Module
22	BLIND_CONTROL	1.2	16.10.07	Ignore angle when T_ANGLE < T#100ms
22	BLIND_SHADE	1.0	15.10.07	New Module
22	CRC_GEN	1.2	09.10.07	Added init values for start value and final XOR
22	REVERSE	1.0	09.10.07	added reflect_in and reflect_out feature
22	SET_DT	1.3	08.10.07	New Module
22	DAY_OF_MONTH	1.3	08.10.07	deleted unused variables count and leap
22	_ARRAY_MEDIAN	1.3	08.10.07	deleted unused variable yr
22	POLYNOM_INT	1.0	08.10.07	deleted unused variables m and temp
22	LINEAR_INT	1.0	07.10.07	New Module
22	BLIND_SECURITY	1.1	06.10.07	Added pos and angle inputs and outputs actions are now configurable
22	BLIND_INPUT	1.1	06.10.07	Renamed output ps and as to po and ao added click mode
22	BLIND_NIGHT	1.1	06.10.07	added pos and angle inputs and outputs night position and angle can now be configuredany manual operation at night will cancel night operation
22	BLIND_CONTROL	1.1	06.10.07	Integrated blind_actuator into BLIND_CONTROL
22	BLIND_ACTUATOR	1.1	06.10.07	Rearranged outputs
22	RMP_W	1.5	05.10.07	Corrected a problem with pt = 0ms
22	RMP_B	1.5	05.10.07	Corrected a problem with pt = 0ms
22	HEX_TO_DWORD	1.0	02.10.07	New Module
21	DAY_OF_DATE	1.1	01.10.07	Added step7 compatibility
21	MONTH	1.1	01.10.07	New coding for much better performance, no string functions used
21	SET_DT	1.2	01.10.07	Added step7 compatibility
21	SET_DATE	1.2	01.10.07	added step7 compatibility, now call function set_date
21	TIMER_1	1.5	01.10.07	Corrected a problem , timer would not generate a pulse over midnight
21	DAY_OF_MONTH	1.2	01.10.07	Corrected an error in algorithm.
21	LEAP_YEAR	1.2	01.10.07	Improved code for higher performance
21	DAY_OF_YEAR	1.1	01.10.07	Added compatibility to step7
21	YEAR	1.1	01.10.07	Corrected error and added compatibility to Step7
21	GLOBAL CONSTANTS		01.10.07	Added Setup Parameter STEP7
21	BLIND_NIGHT	1.0	30.09.07	New Module
21	BLIND_SHADER	1.0	30.09.07	New Module
21	BLIND_INPUT	1.0	29.09.07	New Module
21	BLIND_CONTROL	1.0	29.09.07	New Module
21	BLIND_SECURITY	1.0	29.09.07	New Module
21	BLIND_ACTUATOR	1.0	29.09.07	New Module
21	INTERLOCK	1.0	28.09.07	New Module
21	RMP_W	1.4	28.09.07	Added output high and low, coorrected error while Busy was displayed when en was false inprooved performance and timing accuracy added init statements to avoid wrong startup conditions

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				Added output high and low, coorrected error while Busy was displayed when en was false improved performance and timing accuracy
21	RMP_B	1.4	28.09.07	added init statements to avoid wrong startup conditions
				chagned code to use array_sort
21	_ARRAY_MEDIAN	1.2	22.09.07	Improved performance
21	_ARRAY_SORT	1.2	22.09.07	new algorithm for better performance
21	PERIOD	1.1	20.09.07	corrected a problem with leap year
20	SET_DT	1.1	19.09.07	Corrected leap year calculation
20	SET_DATE	1.1	19.09.07	Corrected leap year calculation
20	PERIOD	1.0	19.09.07	New Module
20	DCF77	1.3	17.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	ESR_MON_X8	1.1	17.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	ESR_MON_R4	1.1	17.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	ESR_MON_B8	1.1	17.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	FT_RMP	1.3	17.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	TP_R	1.1	17.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	SEQUENCE_8	1.4	17.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	SEQUENCE_4	1.4	17.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	GEN_SQ	1.1	17.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	CLK_PULSE	1.1	17.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	CLK_PRG	1.2	17.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	CLK_8MS	1.1	17.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	CLK_4MS	1.1	17.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	CLK_2MS	1.1	17.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	CLK_1MS	1.1	17.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	SH_2	1.3	17.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	SH_1	1.2	17.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	RMP_W	1.3	17.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	RMP_SOFT	1.2	17.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	RMP_B	1.3	17.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	GEN_SQ	1.2	17.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	GEN_SIN	1.1	17.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	GEN_RMP	1.1	17.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	DAYS_DELTA	1.1	16.09.07	Corrected an error in algorithm
20	DAY_OF_DATE	1.0	16.09.07	New Module
20	METER_STAT	1.0	16.09.07	New Module
20	METER	1.0	16.09.07	New Module
20	GEN_RDM	1.1	16.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	ONTIME	1.3	16.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	M_TX	1.2	16.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	M_T	1.2	16.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	M_D	1.1	16.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	FT_TN64	1.1	16.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	FT_TN16	1.1	16.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	FT_PT2	1.2	16.09.07	Deleted unused code section for system time reading and unnecessary startup code
20	FT_PT1	1.4	16.09.07	Replaced Time() with T_PLC_US for compatibility and better internal resolution
20	CYCLE_TIME	1.2	16.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	FT_INT	1.3	15.09.07	Replaced Time() with T_PLC_US for compatibility and better internal resolution
20	FT_DERIV	1.2	15.09.07	Replaced Time() with T_PLC_US for compatibility and better internal resolution
20	FT_PROFILE	1.1	15.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	TEMP_EXT	1.1	15.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	TIMER_2	1.2	15.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	TIMER_1	1.4	15.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	SWITCH_I	1.1	15.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	PULSE_LENGTH	1.1	15.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	DIMM_I	1.3	15.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	CLICK	1.2	15.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	ACTUATOR_UD	1.2	15.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	ACTUATOR_PUMP	1.2	15.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	ACTUATOR_3P	1.3	15.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	ACTUATOR_2P	1.3	15.09.07	Replaced Time() with T_PLC_MS for compatibility and performance reasons
20	SHL1	1.0	14.09.07	New Module
20	SHR1	1.0	14.09.07	New Module
20	TOGGLE	1.0	13.09.07	New Module
20	T_PLC_US	1.0	14.09.07	New Module

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20	T_PLC_MS	1.0	14.09.07	New Module
20	AOUT	1.2	13.09.07	changed code to avoid warning under codesys 2.3.8.1
20	AIN	1.3	13.09.07	changed code to avoid warning under codesys 2.3.8.1
20	RES_NI	1.1	13.09.07	Changed coding for better performance
20	DEW_TEMP	1.2	13.09.07	Changed algorithm to improve performance
20	DEW_CON	1.1	13.09.07	Changed algorithm to improve performance
20	WATER_DENSITY	1.0	13.09.07	New Module
20	DWORD_TO_STRH	1.1	11.09.07	changed coding for compatibility with twincat, under twincat concat cannot have a function as argument.
				changed coding for compatibility with twincat, under twincat concat cannot have a function as argument.
20	BYTE_TO_STRH	1.1	11.09.07	changed coding to avoid a compiler warning under twincat.
20	SET_TOD	1.1	11.09.07	deleted unused variables k and old_code
20	MATRIX	1.1	11.09.07	Deleted unused variable i
20	CRC_GEN	1.1	11.09.07	Deleted unused variable i
20	CHECK_PARITY	1.1	10.09.07	Changed algorithm to improve performance
20	PARITY	1.1	10.09.07	Changed algorithm to improve performance
20	TEMP_SII	1.2	10.09.07	Reduced accuracy to 0.01 degree to shorten execution time and corrected an error in formula
				Reduced accuracy to 0.01 degree to shorten execution time
20	TEMP_PT	1.2	10.09.07	Reduced accuracy to 0.01 degree to shorten execution time
20	TEMP_NI	1.2	10.09.07	Reduced accuracy to 0.01 degree to shorten execution time
20	BIT_COUNT	1.1	10.09.07	Improved performance
20	SH_2	1.2	10.09.07	Corrected an error while index would be out of array if N = 0;
20	D_TRIG	1.0	04.09.07	New Module
18	ONTIME	1.2	02.08.07	Corrected an error while ontime would stop counting time after a certain time is reached
17	LTIME_TO_UTC	1.0	05.07.07	New Module
17	BIT_COUNT	1.0	05.07.07	New Module
17	MATRIX	1.0	10.06.07	New Module
17	CRC_CHECK	1.0	09.06.07	New Module
17	CRC_GEN	1.0	09.06.07	New Module
17	BYTE_TO_STRB	1.0	09.06.07	New Module
17	BYTE_TO_STRH	1.0	09.06.07	New Module
17	DWORD_TO_STRB	1.0	09.06.07	New Module
17	DWORD_TO_STRH	1.0	09.06.07	New Module
17	CLICK	1.1	18.05.07	Corrected an error while Click would stop working when t_reconfig expired
16	PRESSURE	1.0	27.03.07	New Module
16	ARRAY_SDV	1.0	01.04.07	New Module
16	LTIME	1.3	18.03.07	changed code, dst would not work during first cycle.
16	DEW_TEMP	1.1	29.04.07	Changed code when RH = 0, not to execute log(0)
16	GLOBAL CONSTANTS		02.05.07	Added UG, universal gas constant
16	ARRAY_VAR	1.0	01.04.07	New Module
16	TIMER_1	1.3	14.04.07	Corrected an error while output would not go active when start year = 1970
				increased accuracy by checking every 100ms
16	DATE_ADD	1.1	12.04.07	added init section to compensate for timer overflow at startup
16				corrected an error while date would be incorrect when year = 0
				Corrected wrong measurements during startup
16	M_TX	1.1	02.05.07	added output ET (elapsed time)
				added Reset input
16	LENGTH	1.0	27.03.07	added tmax input
16	ASTRO	1.0	27.03.07	New Module
16	ENERGY	1.0	27.03.07	New Module
16	DEW_CON	1.0	29.04.07	New Module
16	CYCLE_TIME	1.2	10.03.07	changed initialization of ct_min to t#10h for compatibility with siemens s7
16	SPEED	1.0	27.03.07	New Module
16	M_D	1.0	02.05.07	New Module
16	M_T	1.1	02.05.07	Added init variable to avoid unreasonable result if falling edge occurs first
				added ET (elapsed Time) output
16	DEW_RH	1.0	29.04.07	added reset input
16	SUN_TIME	1.2	17.04.07	New Module
15	MUL_ADD	1.0	08.02.07	corrected error while sun:midday would not be corrected for longitude.
15	CLK_PRG	1.1	25.02.07	New Module
				rewritten module in ST for better performance
15	DEC_8	1.1	03.03.07	PT can now be changed during runtime
15	FT_RMP	1.2	12.02.07	rewritten module in ST for better performance
15	GEN_RDM	1.0	07.02.07	Added INIT to prevent erroneous startup conditions
				New Module

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15	FT_PID	1.3	03.03.07	Added default values for inputs KP, TN, TV, LIMIT_H and LIMIT_L
15	CEIL	1.0	07.02.07	New Module
15	FT_PT1	1.3	27.02.07	Output will be set to input during initial startup
15	CLICK	1.0	15.02.07	New Module „click“
15	PULSE_LENGTH	1.0	15.02.07	New Module
15	FACT	1.1	04.03.07	Changed a critical erro while loop was indefinite
15	GEN_RMP	1.0	03.03.07	New Module
15	FLOOR	1.0	07.02.07	New Module
15	DEAD_ZONE_2	1.0	12.02.07	New Module
15	SIGN_I	1.1	03.03.07	changed method of function for better compatibility to other systems Increased accuracy of PI2 renamed T0 to TK for better compatibility with other Libraries
15	GLOBAL CONSTANTS		03.03.07	renamed C to C0 for better compatibility
15	WCT	1.0	07.02.07	New Module
15	HOLIDAYEAT_INDEX	1.1	27.02.07	deleted unused variables rh3 and T3
15	PWM_DC	1.1	25.02.07	rewritten moule in ST for better performance
15	F_LAMP	1.1	08.02.07	New Module
15	HOLIDAY	1.1	27.02.07	deleted unused variable init
15	ONTIME	1.1	22.02.07	Changed VAR RETAIN PERSISTENT to VAR RETAIN for better compatibility with controllers without file system like WAGO's LON controllers
15	TIMER_2	1.1	22.02.07	Changed VAR RETAIN PERSISTENT to VAR RETAIN for better compatibility with controllers without file system like WAGO's LON controllers
15	_ARRAY_MEDIAN	1.1	03.03.07	Corrected an error in median calculation
15	DEC_4	1.1	03.03.07	rewritten moule in ST for better performance
15	GEN_SQR	1.1	12.02.07	Added default value for dc = 0.5
15	SWITCH_X	1.0	15.02.07	New Module
15	CALIBRATE	1.2	22.02.07	Chaged VAR RETAIN PERSISTENT to VAR RETAIN for better compatibility with controllers without file system
15	DEC_2	1.1	03.03.07	rewritten moule in ST for better performance
15	FT_PROFILE	1.0	27.02.07	New Module
15	INC	1.0	07.02.07	New Module
15	PWM_PW	1.1	25.02.07	rewritten moule in ST for better performance
15	DEAD_ZONE	1.0	12.02.07	New Module
15	BAR_GRAPH	1.0	22.02.07	New Module
15	BOILER	1.0	27.02.07	New Module
15	DCF_77	1.2	26.02.07	Chaqnged statements to avoid a compiler error if wago 16 bit lon controller is used
14	MUX_R2	1.0	19.01.07	New Module
14	DIMM_I	1.1	22.01.07	Deleted unused structure „ramp“
14	F_TO_OM	1.0	22.01.07	New Module
14	ESR_MON_X8	1.0	26.01.07	New Module
14	ASINH	1.0	12.01.07	New Module
14	DAY_OF_MONTH	1.1	22.01.07	deleted unused variable day_in_year and day_in_year_begin
14	Heat_Temp	1.0	27.01.07	New Module
14	DIMM_I	1.2	02.02.07	Changed method to read time() only at the beginning of the module for better consistency when a higher priority task interrupts „dimm_i“ added code in case structure for state 1 to allow for short one cycle long input pulses
14	SEQUENCE_4	1.3	17.01.07	Changed output fault to ESR compatible output STATUS added stop on error functionality default for inputs IN is now TRUE renamed variable state to step
14	SH_2	1.1	20.01.07	Added input N to specify the amount of samples for calculations
14	DWORD_TO_STRF	1.0	16.01.07	added output trig New Module
14	_ARRAY_INIT	1.1	06.01.07	Changed type of function to BOOL function now returns TRUE
14	SUN_POS	1.0	01.02.07	New Module
14	SCALE_B4	1.1	19.01.07	Changed outputs to REAL to avoid Integer overflow added input offset for better cascading
14	SCALE_X8	1.0	19.01.07	changed from FB to FUNCTION
14	STAIR2	1.0	28.01.07	New Module
14	HYST_3	1.0	22.01.07	New Module
14	GEN_SIN	1.0	22.01.07	New Module
14	SIGN_R	1.1	19.01.07	changed method of function for better compatibility to other systems
14	RMP_W	1.2	22.01.07	Deleted unused variable step
14	KMH_TO_MS	1.0	04.02.07	New Module

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14	ACTUATOR_2P	1.2	17.01.07	Deleted unused variable time_overflow
14	ACTUATOR_3P	1.2	23.01.07	Deleted unused variables „force_on“ and „force_off“
14	FT_PT2	1.1	15.01.07	Improved algorithm for higher accuracy and speed
14	SENSOR_INT	1.1	20.01.07	Deleted input R0 which was not used
14	SQRTN	1.0	12.01.07	New Module
14	BURNER	1.3	31.12.06	Added status output to be ESR compatible
14	F_LIN2	1.0	01.01.07	New Module
14	FT_TN16	1.0	01.01.07	New Module
14	FT_PID	1.2	03.01.07	Added int_band for structural adaptation of PID
14	BYTE_OF_DWORD	1.0	17.01.07	New Module
				Renamed module to AOUT
				changed output to hold 32 bits max
14	AOUT	1.1	18.01.07	corrected an error with sign bit
14	SUN_TIME	1.1	20.01.07	deleted unused variables sun_rineR and sun_setR
14	SH	1.1	16.01.07	Added output trig
14	MAX_3	1.0	01.01.07	New Module
14	OM_TO_F	1.0	22.01.07	New Module
14	MIN_3	1.0	01.01.07	New Module
				Corrected an error with falling edge errors
14	FT_PT1	1.2	03.01.07	added faktor K
				Changed output fault to ESR compatible output STATUS
				added stop on error functionality
				default for inputs IN is now TRUE
14	SEQUENCE_8	1.3	17.01.07	renamed variable state to step
14	MULTI_IN	1.1	01.01.07	Changed midr to mid3 funktion for better performance
14	STAIR	1.0	28.01.07	New Module
14	ACOSH	1.0	12.01.07	New Module
14	BURNER	1.2	31.12.06	Added Oil_Temp input
				Added status output
				changed inputs „ton“ and „toff“ to config variables
				added config var „out_return to enable „yup_in“ and y_dn_in“
14	ACTUATOR_UD	1.1	22.01.07	changed manual mode to override automatic mode
14	FT_TN8	1.0	01.01.07	New Module
				Changed outputs to REAL to avoid Integer overflow
				added input offset for better cascading
14	SCALE_B8	1.1	19.01.07	changed from FB to FUNCTION
14	DELAY	1.2	19.01.07	changed reset to load the value of in instead of 0
14	CHR	1.0	16.01.07	New Module
14	SCALE_X4	1.0	19.01.07	New Module
14	RDM	1.0	16.01.07	New Module
				Rewritten moule in ST for better performance
14	LEGIONELLA	1.1	17.01.07	changed error out to ESR compatible Status output
14	ACOTH	1.0	12.01.07	New Module
14	WORK_WEEK	1.1	14.01.07	deleted unused variable yday
14	SCALE_X2	1.0	19.01.07	New Module
14	DCF_77	1.1	02.02.07	change wday and dst outputs when there is no dcf reception
14	FT_IMP	1.1	03.01.07	Added faktor K
14	SH_1	1.1	06.01.07	Added output trig
14	MUX_R4	1.0	19.01.07	New Module
14	DELAY_4	1.1	19.01.07	added variable temp to add 1 delay for q1
14	WORD_OF_DWORD	1.0	17.01.07	New Module
14	_ARRAY_SORT	1.1	06.01.07	Function now returns TRUE
				Changed output „fail“ to „status“
				added output „cycles“
				corrected an error while pre_heat would be active when an error was present
14	BURNER	1.4	22.01.07	added „coil2“ output and „stage2“ input for dual stage burners
14	TEMP_PT	1.1	24.01.07	deleted input R0 which was not used
14	OFFSET_2	1.0	12.01.07	New Module
				Corrected error during startup
14	FT_PT1	1.1	01.01.07	corrected error with very small time values < 10ms
14	COSH	1.0	12.01.07	New Module
14	TANH	1.0	12.01.07	New Module
				Added init code for startup
14	FT_DERIV	1.1	03.01.07	set default for K to 1
14	RMP_SOFT	1.0	22.01.07	New Module
14	ATANH	1.0	12.01.07	New Module

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14	FT_INT	1.2	15.01.07	Added default value for K = 1
14	COTH	1.0	12.01.07	New Module
14	DWORD_OF_BYTE	1.0	24.01.07	New Module
14	MS_TO_KMH	1.0	04.02.07	New Module
14	RMP_B	1.2	17.01.07	Renamed input UD to UP for better usability
14	TEMP_NI	1.1	24.01.07	deleted unused variable step
14	ESR_MON_B8	1.0	26.01.07	deleted input R0 which was not used
14	ESR_MON_R4	1.0	26.01.07	New Module
14	FT_AVG	1.1	07.01.07	Changed reset logic to load buffer with IN instead of 0
14	OFFSET	1.0	12.01.07	added EN input for better flow control
14	EXEC	1.1	06.02.07	added init to avoid erroneous startup conditions
14	AIN	1.2	19.01.07	deleted unused variable CNT
14	LTIME	1.2	17.01.07	Corrected error with COS function
14	BURNER	1.1	31.12.06	divide by 0 wilol return an error now
14	WINDOW	1.1	22.01.07	fixed an error with sign bit.
14	TEMP_SI	1.1	24.01.07	Added input UTC instead of internal SYSRTCGETTIME for better compatibility with different Hardware
14	WORD_OF_BYTE	1.0	24.01.07	corrected an error while DST_ENABLE would not be checked before DST is enabled
14	ESR_COLLECT	1.0	26.01.07	Added Pre heating time and output
14	SINH	1.0	12.01.07	flame monitor before ignition will generate error output now
14	FT_TN64	1.0	01.01.07	added output for Kwh
14	MID_3	1.0	01.01.07	Changed input arrangement
14	SCALE_B2	1.1	19.01.07	deleted input R0 which was not used
13	ACTUATOR_2P	1.1	07.10.06	New Module
13	DATE_ADD	1.0	27.12.06	New Module
13	TEMP_EXT	1.0	06.12.06	New Module
13	K_TO_C	1.0	04.12.06	New Module
13	RMP_B	1.1	15.10.06	Added output busy
13	FIND_NONUM	1.0	06.10.06	New Module
13	FT_PID	1.1	01.12.06	Changed algorithm to „Trapezregel“ for higher accuracy
13	TIMER_1	1.2	25.12.06	Corrected an error with timer overflow
13	TIMER_1	1.1	01.12.06	Corrected an error with pulse width = 0
13	C_TP_K	1.0	04.12.06	New Module
13	CYCLE_TIME	1.1	12.12.06	Added output cycles
13	TRIM	1.0	06.10.06	New Module
13	FINDB_NUM	1.0	06.10.06	New Module
13	MUX_4	1.1	05.10.06	rewritten moule in ST for better performance
13	FT_RMP	1.1	08.10.06	Added UD output
13	FIND_NUM	1.0	06.10.06	New Module
13	ACTUATOR_2P	1.1	07.10.06	changed „on“ and „off“ to „force“ and „force_in“ for better usability.
13	SEQUENCE_4	1.2	01.12.06	corrected an error in sequence logic
13	MUX_2	1.1	05.10.06	added init at startup to prevent erroneous startup conditions
13	UPPERCASE	1.0	06.10.06	rewritten moule in ST for better performance
13	DAYS_DELTA	1.0	27.12.06	New Module
13	FT_INT	1.1	13.12.06	Changed algorithm to „Trapezregel“ for higher accuracy
13	ACTUATOR_3P	1.1	19.10.06	Added code to avoid overrun of the end_switch
13	OSCAT_VERSION	1.0	06.10.06	New Module
13	LOWERCASE	1.0	06.10.06	New Module
13	FINDB	1.0	06.10.06	New Module
13	REAL_TO_STRF	1.0	06.10.06	New Module
13	ACTUATOR_PUMP	1.1	27.12.06	Fixed an error while the pump would run for t_min at startup
13	EASTER	1.0	27.12.06	New Module
13	DIFFER	1.0	08.10.06	New Module
13	FINDB_NONUM	1.0	06.10.06	New Module

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				corrected an error in sequence logic
13	SEQUENCE_8	1.2	01.12.06	added init at startup to prevent erroneous startup conditions
13	STATUS_TO_ESR	1.0	06.10.06	New Module
12	SECOND	1.1	02.10.06	Changed name of input to ITOD
12	RMP_W	1.1	04.10.06	added the busy output which signals that the ramp is running.
12	ARRAY_SUM	1.0	02.10.06	New Module
12	ARRAY_TREND	1.0	02.10.06	New Module
12	LTIME	1.1	02.10.06	corrected an error where dst would be delayed by 0.1second
12	LEAP_OF_DATE	1.1	02.10.06	Corrected an error while year 1970 would not work
12	CHECK_PARITY	1.0	01.10.06	New Module
12	DELAY	1.1	01.10.06	Corrected an error in buffer management
12	ARRAY_MIN	1.0	02.10.06	New Module
12	ARRAY_MAX	1.0	02.10.06	New Module
				Added output any
12	SEQUENCE_4	1.1	01.10.06	corrected delay to occur aftzer event and not before
12	LEAP_YEAR	1.1	02.10.06	Corrected an error while year 1970 would not work
12	ARRAY_AVG	1.0	02.10.06	New Module
12	ARRAY_SPR	1.0	02.10.06	New Module
12	LOCAL_TIME	1.1	02.10.06	Corrected an error while DST would be calculated wrong
12	HOURL	1.1	02.10.06	Changed name of input to ITOD
12	MULTIME	1.0	02.10.06	New Module
12	EVEN	1.0	01.10.06	New Module
12	DEW_TEMP	1.0	02.10.06	New Module
12	MINUTE	1.1	02.10.06	Changed name of input to ITOD
				Added output any
12	SEQUENCE_8	1.1	01.10.06	corrected delay to occur aftzer event and not before
11	SH_2	1.0	01.09.06	New Module
11	SH_T	1.0	01.09.06	New Module
11	FT_MIN_MAX	1.0	01.09.06	New Module
11	PARITY	1.0	01.09.06	New Module
11	F_LIN	1.0	01.09.06	New Module
11	F_QUAD	1.0	01.09.06	New Module
11	AIN	1.1	18.08.06	fixed an error with low value negative and high value 0.
11	ROUND	1.0	01.09.06	New Module
11	F_POWER	1.0	01.09.06	New Module
11	WEEKDAY	1.1	21.08.06	Coorected an error in formula
10	TP_R	1.0	04.08.06	New Module
10	SET_DT	1.0	04.08.06	New Module
10	M_TX	1.0	04.08.06	New Module
10	RAD	1.0	04.08.06	New Module
10	INC_DEC	1.0	04.08.06	New Module
10	MINUTE_TO_TIME	1.0	04.08.06	New Module
10	SHR_4UDE	1.0	04.08.06	New Module
10	DAY_OF_YEAR	1.0	04.08.06	New Module
10	HYST_2	1.0	04.08.06	New Module
10	MUX_4	1.0	04.08.06	New Module
10	SECOND_TO_TIME	1.0	04.08.06	New Module
10	FF_D2E	1.0	04.08.06	New Module
10	CLK_4MS	1.0	04.08.06	New Module
10	DST	1.0	04.08.06	New Module
10	FRACT	1.0	04.08.06	New Module
10	SET_TOD	1.0	04.08.06	New Module
10	FF_D4E	1.0	04.08.06	New Module
10	CLK_DIV	1.0	04.08.06	New Module
10	CLK_2MS	1.0	04.08.06	New Module
10	PT_TO_F	1.0	04.08.06	New Module
10	CLK_PULSE	1.0	04.08.06	New Module
10	M_T	1.0	04.08.06	New Module
10	DAY_OF_MONTH	1.0	04.08.06	New Module
10	DAY_TO_TIME	1.0	04.08.06	New Module
10	RES_PT	1.1	04.08.06	New Module
10	FIFO_16	1.0	04.08.06	New Module
10	STACK_32	1.0	04.08.06	New Module
10	F_TO_PT	1.0	04.08.06	New Module
10	RES_NI	1.0	04.08.06	New Module

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10	CLK_1MS	1.0	04.08.06	New Module
10	MUX_2	1.0	04.08.06	New Module
10	BIT_OF_DWORD	1.0	04.08.06	New Module
10	FF_JKE	1.0	04.08.06	New Module
10	LTCH_4	1.0	04.08.06	New Module
10	GEN_SQ	1.0	04.08.06	New Module
10	GEN_BIT	1.0	04.08.06	New Module
10	SHR_8UDE	1.0	04.08.06	New Module
10	HOUR	1.0	04.08.06	New Module
10	SHR_4E	1.0	04.08.06	New Module
10	B_TRIG	1.0	04.08.06	New Module
10	BYTE_OF_BIT	1.0	04.08.06	New Module
10	BYTE_TO_BIT	1.0	04.08.06	New Module
10	SHR_8PLE	1.0	04.08.06	New Module
10	C_TO_F	1.0	04.08.06	New Module
10	FF_DRE	1.0	04.08.06	New Module
10	STACK_16	1.0	04.08.06	New Module
10	SET_DATE	1.0	04.08.06	New Module
10	DEG	1.0	04.08.06	New Module
10	SWITCH_I	1.0	04.08.06	New Module
10	HOUR_TO_TOD	1.0	04.08.06	New Module
10	YEAR	1.0	04.08.06	New Module
10	HYST_1	1.0	04.08.06	New Module
10	MONTH	1.0	04.08.06	New Module
10	HOUR_TO_TIME	1.0	04.08.06	New Module
10	LTCH	1.0	04.08.06	New Module
10	CLK_8MS	1.0	04.08.06	New Module
10	RES_SI	1.0	04.08.06	New Module
10	F_TO_C	1.0	04.08.06	New Module
10	FIFO_32	1.0	04.08.06	New Module